

These results may be extended to non-renormalizable theories.³ In such theories the first term $[\Phi^\dagger e^{-V} \Phi]_D$ in Eq. (27.6.1) is replaced with the D -term of an arbitrary real gauge-invariant scalar function of $\Phi^\dagger$, Φ , V , and their superderivatives and spacetime derivatives, while the second and third terms in Eq. (27.6.1) are replaced with the $\mathcal{F}$ -term of an arbitrary globally gauge-invariant scalar function $f(\Phi, W)$ of Φ_n and W_α . It has been shown that, to all orders of perturbation theory, the function $f_\lambda(\Phi, W)$ appearing in the $\mathcal{F}$ -term of the Wilsonian Lagrangian is the same as $f(\Phi, W)$, except for the one-loop renormalization of the term quadratic in W .

27.7 Soft Supersymmetry Breaking*

We will see in the next chapter that, even if supersymmetry is an exact symmetry of the action, the spontaneous breakdown of supersymmetry at very high energy can produce superrenormalizable terms that violate supersymmetry conservation in the effective action that describes physics at lower energies. These superrenormalizable terms may explain the lack of supersymmetry observed in phenomena at accessible energies. In this section we will consider the radiative corrections that can be produced by such supersymmetry-breaking superrenormalizable terms, in part to see whether this provides a criterion for including or rejecting such terms in supersymmetric versions of the standard model.

The sign of supersymmetry breaking is the appearance of expectation values of D -terms of general superfields or $\mathcal{F}$ -terms of chiral superfields. Any operator $\epsilon \mathcal{O}$ in the Lagrangian density that breaks supersymmetry can be written in a supersymmetric form, as a D -term

$$\epsilon \mathcal{O} = [Z S]_D, \quad (27.7.1)$$

where S is a non-chiral superfield that has $\mathcal{O}$ as its C -term, and Z is a non-chiral external superfield whose only non-vanishing component is $[Z]_D = \epsilon$. Some but not all operators $\epsilon \mathcal{O}$ that break supersymmetry can also be written as $\mathcal{F}$ -terms,

$$\epsilon \mathcal{O} = [\Omega O]_{\mathcal{F}}, \quad (27.7.2)$$

or their adjoints, where O is the left-chiral superfield whose $\mathcal{F}$ -term is $\mathcal{O}$ and Ω is an external left-chiral superfield whose only non-vanishing component is $[\Omega]_{\mathcal{F}} = \epsilon$. We can count the order in ϵ in which a given

* This section lies somewhat out of the book's main line of development and may be omitted in a first reading.